



ID number _____ Date _____ Group _____

Signature

- 1) Which of the following options evaluates correctly $(g \circ f)(x)$
a) $g(f(x))$ b) $g(x)f(x)$ c) $f(g(x))$ d) $f \cdot g \cdot x$

- 2) Which of the following functions models exponential decay.
a) $m(t) = -Ae^{-rt}$ b) $m(t) = -Ae^{r-t}$ c) $m(t) = Ae^{-r+t}$ d) $m(t) = Ae^{-rt}$

- 3) Considering $f(x) = (x + 1)^2 + 1$. What would the new function be like after shifting 5 units to the left and 3 units down?
a) $f(x) = (x + 6)^2 - 2$ b) $f(x) = (x + 2)^2 - 4$ c) $f(x) = (x + 4)^2 - 2$ d) $f(x) = (x + 6)^2 - 4$

- 4) Given the function $f(x) = -2\sqrt{x^2 - 3} + 5$, which of the following represents its domain?
a) $-\infty < x \leq 0$ b) $-\infty < x \leq 4$ c) $-\infty < x < \infty$ d) $2 \leq x < \infty$

- 1) Solve the following equations. Simplify the answer as much as possible and check that all solutions satisfies the original equation. **(6 points each equation)**

a. $\log(3-x) + \log(x-9) = 0$

$$\text{b. } -2^{x-3} + 6^{x+2} = 0$$

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- 2) For the functions $f(x) = \frac{x+2}{5x}$ and $g(x) = \frac{8}{x}$ find the indicated composition and reduce the expression as much as possible, **(6 points each operation)**

a. $(g \circ f)(x)$	b. $(f + g)(x)$

- 3) Use logarithmic properties to condense the following expression $3 \log_8 y + \frac{1}{4}(\log_8 w + 4 \log_8 x)$. **(6 points.)**

Section 3: Function analysis (12 points)

- 1) Find the inverse function of $f(x) = \frac{2x-10}{3x-4}$. Reduce the expression as much as possible and proof your answer by evaluating the function composition $(f \circ f^{-1})(x) = x$. **(6 points)**

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- 2) Find the roots of the function $f(x) = \frac{-x^2 - x + 2}{x + 2}$. **(6 points)**

Section 4: Modeling with math (30 points)

- 1) Hydrogen potential (pH) is a number used to describe the acidity or basicity of a chemical substance and is defined by the equation $pH = -\log(H^+)$, where H^+ measures the concentration of hydrogen ions moles per liter.
- i. Determine the pH of a coke if $H^+ = 1.4 \times 10^{-3}$. **(6 points)**
 - ii. What is the hydrogen ions concentration of the stomach acid with pH of 1.6?. **(6 points)**

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- 2) The half-life of Radium-266 is 1600 years. Suppose that there is a sample of 12mg. Recalling that the mass decays exponentially, fulfill the following points
- i) Write the function that models the remaining mass after t years. **(6 points)**
 - ii) How much mass will remain after 4050 years? **(6 points)**
 - iii) How much time will it take for the sample to be only a quarter of its initial value?. **(6 points)**

Section 5: Conic Sections (4 points)

- 1) Identify the conic section given the following general equation $9 - 6x + x^2 - 4y^2 = 4$. **(4 points.)**